

ADEMP pre-registration: 07-gompertz-evaluation

2026-05-07 16:55 PDT

This document pre-registers the simulation programme that underlies `analysis/report/07-gompertz-evaluation/`. ADEMP structure follows Morris, White and Crowther (2019).

Common infrastructure

Trial design. Hendrickson hybrid OL+BDC design, 16-week total trial duration, 8-week open-label active drug + 4-week blinded discontinuation + 4-week crossover.

Sample sizes. $N \in \{35, 70, 100\}$.

Drug context. Prazosin for PTSD nightmares, calibrated to the Hendrickson 2020 reference parameters at the saturating asymptote.

Replicates. $n_{\text{reps}} = 1000$ per cell for power and bias estimands; $n_{\text{reps}} = 5000$ for type I error cells.

Random-number control. Per-replicate seed via cell-descriptor hash; base seed in `01-base-seed.txt`.

Reporting standard. Morris-White-Crowther ADEMP with MCSE columns for every reported performance measure.

Calibration to common effect size. All four DGP families share the same marginal Cohen’s-style standardised effect size at week 8 in the open-label phase, computed as the marginal on-drug $\text{Cor}(B, BR)$ in the saturated regime. Each family’s parameters are fitted to match this effect size before the production grid runs, via the calibration sub-study described in Study 0 below.

Study 0. Effect-size calibration sub-study

A. Fit the parameters of each of the four DGP families (Gompertz, symmetric logistic, hyperbolic-tangent, piecewise-linear breakpoint) so that all four produce the same marginal on-drug $\text{Cor}(B, BR)$ at week 8.

D. Single-component BR-only generative model under each of the four families, with population-level participant heterogeneity but no PB or TV component (those are added in Study 1).

E. The set of family-specific parameters that produces matched marginal effect size.

M. Numerical optimisation via `optim`, with an analytic Jacobian where available.

P. Convergence to the calibration target within $|d| < 0.01$ on Cohen’s-style effect size; reproducibility across random seeds.

Study 1. Power, bias, and type I error across families

A. Quantify how the standard linear-mixed analysis’s inferential properties (power, type I error, bias) depend on the DGP family used to generate the data.

D. Four DGP families on the BR factor only:

1. *Modified Gompertz* (the `pmsimstats` default): $BR(t) = m \exp(-d \exp(-rt))$ with vertical-offset adjustment.
2. *Symmetric logistic*: $BR(t) = m / (1 + \exp(-r(t - t_0)))$.
3. *Hyperbolic-tangent*: $BR(t) = (m/2)(1 + \tanh(r(t - t_0)))$.
4. *Piecewise-linear breakpoint*: $BR(t) = m \cdot \min(1, t/t_0)$.

PB and TV components are held at their Hendrickson reference specifications (`modGompertz` with calibrated parameters), so the family axis varies BR alone. Each family is calibrated to the matched marginal effect size from Study 0.

E. Primary estimand: $\hat{\beta}_{bm:D}$ from the linear- mixed analysis. Secondary estimand: posterior class entropy is not applicable here (single-component DGPs), but the empirical SD of $\hat{\beta}_{bm:D}$ across replicates is reported as a finite-sample efficiency measure.

M. Single analysis: linear mixed-effects model $Sx \sim bm + t + Dbc + bm:Dbc + (1 | ptID)$ with `corCAR1(form = ~t|ptID)`.

P. Power for $H_0 : \beta_{bm:D} = 0$, type I error under $\beta_{bm:D} = 0$, bias of $\hat{\beta}_{bm:D}$, empirical SE across replicates, model SE, nominal-95% CI coverage, convergence rate. MCSE attached to every estimate.

Pre-registered prediction. Power varies less than $\pm 5\%$ across the four families at fixed sample size and effect size, provided each is calibrated to the matched marginal effect size. Type I error remains within $\pm 1\%$ of nominal across families. Bias is small at all four families when the analysis model's continuous-decay carryover specification matches the DGP's saturating-shape regime.

Total cells. 4 families $\times$ 3N $\times$ 3 effect sizes $\times$ $4t_{1/2,DGP} = 144$ cells.

Study 2. Sloppiness and parameter identifiability

A. Characterise the parameter-redundancy structure (sloppiness) of the modified Gompertz form fitted to short N-of-1 trajectories.

D. Single Gompertz BR generative model at the Hendrickson reference; eigenvalue analysis of the parameter Hessian at the maximum-likelihood point, computed analytically.

E. The set of eigenvalues of the parameter Hessian, ordered by magnitude. Following @jagadeesan2023sloppiness, the sloppiness is characterised by the ratio of the largest to the smallest eigenvalue.

M. Numerical Hessian computation at the MLE for each replicate; analytical comparison to the unified-Richards reparameterisation of @tjorve2010richards.

P. Mean and variance of each eigenvalue across replicates; the sloppiness ratio; the sample-size scaling of the smallest eigenvalue.

Deliverables

- Driver scripts at `analysis/scripts/gompertz-evaluation/`.
- ADEMP results tables per study with MCSE columns.
- Figures: power/bias/type-I plots faceted by DGP family; sloppiness eigenvalue spectrum.

Deviations log

`02-deviations-log.md` records departures from the cells, estimands, methods, or performance measures pre-registered above.

Source: `~/prj/alz/10-pmsimstats-ng/pmsimstats-ng/analysis/scripts/gompertz-evaluation/00-ademp-pre-registration.md`